Errata for *Chapter 1 — Rings and Ideals*

October 27, 2026

Exercise numbers below refer to the order of the `exercise` environments in the original file, which coincides with Atiyah–MacDonald, Chapter 1, Exercises 1–28. For each item, **Issue** describes the problem and **Correction** gives a fix. Exercises which are only sketches or are unfinished are collected in the last two sections.

Summary of serious mathematical errors

- **Ex. 2(iii)(a)**: the displayed system of equations is wrong, and the conclusion $fb_m^n = 0$ does not produce a nonzero annihilator.
- **Ex. 2(iv)**: the proof of "f, g primitive $\Rightarrow$ fg primitive" is not valid.
- **Ex. 4**: $A[x]$ need not have Krull dimension 0; the asserted statement "every prime ideal of $A[x]$ is maximal" is false.
- **Ex. 5(iii)(b)**: the claim that an arbitrary ideal equals the full preimage of its constant-term ideal is false, so the classification of maximal ideals is not proved.
- **Ex. 6**: the logical reduction is incorrect and the proof is unfinished.
- **Ex. 8**: the nilradical is not a prime ideal in general, so it cannot be used as a minimal prime.
- **Ex. 13**: the minimal-degree argument is not written correctly.
- **Ex. 14(ii)**: the proof that a maximal element of Σ is prime is invalid.
- **Ex. 16**: the zero ideal is missing from $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathbb{C}[x]$, and $\operatorname{Spec}\mathbb{R}[x]$.
- **Ex. 17(vi),(vii)**: the radical is confused with the ideal itself, and the quasi-compactness argument in (vii) is wrong.
- **Ex. 18(iv)**: $V(\mathfrak{p}_x)$ is closed but is not a neighbourhood of x in general; the T_0 proof is not valid as written.
- **Ex. 19**: the category-theoretic argument does not prove irreducibility of $\operatorname{Spec}(A)$.
- **Ex. 21(v)**: the chain of equivalences is not justified; the correct criterion is $\ker\phi \subseteq \operatorname{Nil}(A)$.
- **Ex. 23(iii),(iv)**: the proof that clopen sets are basic is wrong, and $X \setminus \{q\}$ is not closed.
- **Ex. 24**: the associativity computation for addition and the absorption computation are incorrect.
- **Ex. 26**: the implication "$f \in \mathfrak{m}_y \Rightarrow f(y) \neq 0$" has the wrong sign.
- **Ex. 27**: the argument uses the weak Nullstellensatz, which is exactly what is not yet available at this point.

- **Ex. 28**: the crucial verification that the constructed map lands in Y is missing, and injectivity is not proved.

Exercise 1

Issue. The identity

$$(1+x)(1-x+x^2-\cdots+(-1)^{r-1}x^{r-1}) = 1-x^r$$

is not correct in general. The correct identity is

$$(1+x)(1-x+x^2-\cdots+(-1)^{r-1}x^{r-1}) = 1-(-x)^r.$$

Since $x^r = 0$, both $1-x^r$ and $1-(-x)^r$ equal 1, so the final conclusion is correct. The exercise also asks to deduce that the sum of a unit and a nilpotent element is a unit; this deduction is not written.

Correction. Write

$$(1+x)(1-x+\cdots+(-1)^{r-1}x^{r-1}) = 1-(-x)^r = 1.$$

For the deduction, if u is a unit and $y^n = 0$, then

$$u+y = u(1+u^{-1}y),$$

and $(u^{-1}y)^n = 0$. By the first part, $1+u^{-1}y$ is a unit, hence $u+y$ is a unit.

Exercise 2

(i)(a)

Correct. The displayed system

$$a_n b_m = 0, \quad a_n^2 b_{m-1} = 0, \quad \ldots, \quad a_n^{m+1} b_0 = 0$$

is obtained by multiplying the coefficients of $x^{n+m}, x^{n+m-1}, \ldots$ by suitable powers of a_n. Since $a_0 b_0 = 1$, we get

$$a_n^{m+1} = a_n^{m+1} b_0 a_0 = 0.$$

Then $f - a_n x^n$ is a unit by Exercise 1, and iteration gives the nilpotence of $a_{n-1}, \ldots, a_1$.

(i)(b)

Correct, except for the typo $a_n s^n$ in line 87 of the original; it should be $a_n x^n$.

(ii)(a)

The idea is correct, but one step needs justification: from the nilpotence of

$$xh = a_1 x + a_2 x^2 + \cdots + a_n x^n$$

one cannot immediately conclude that $h = a_1 + a_2 x + \cdots + a_n x^{n-1}$ is nilpotent without using that x is a non-zero-divisor in $A[x]$. Indeed, if $(xh)^k = 0$, then $x^k h^k = 0$, hence $h^k = 0$.

(ii)(b)

The word "obviously" hides a short argument. If each a_i is nilpotent, then each $a_i x^i$ is nilpotent, and a finite sum of nilpotent elements is nilpotent. Hence f is nilpotent.

(iii)(a): incorrect system of equations

Issue. The displayed system

$$a_n b_m = 0, \quad a_{n-1} b_m = 0, \quad a_{n-2} b_m^2 = 0, \quad \dots, \quad a_1 b_m^{n-1} = 0, \quad a_0 b_m^n = 0$$

is not correct. For example, comparing the coefficient of x^{n+m-1} gives

$$a_{n-1} b_m + a_n b_{m-1} = 0.$$

Multiplying by b_m and using $a_n b_m = 0$ gives

$$a_{n-1} b_m^2 = 0,$$

not $a_{n-1} b_m = 0$. In general, without using minimality of m, one only gets

$$a_{n-r} b_m^{r+1} = 0 \qquad (0 \le r \le n),$$

hence $f b_m^{n+1} = 0$. But $b_m \neq 0$ does not imply $b_m^{n+1} \neq 0$, so this does not give the required nonzero annihilator.

A concrete counterexample to the displayed system is the following. Let

$$A = k[x, y]/(x^2, y^2), \qquad f = x + yX, \qquad g = -x + yX.$$

Then

$$fg = -x^2 + xyX - xyX + y^2 X^2 = 0,$$

but with $a_0 = x, a_1 = y, b_0 = -x, b_1 = y$ we have

$$a_0 b_1 = xy \neq 0.$$

Thus the equation $a_0 b_m^n = 0$ is false without further hypotheses.

Correction. Use the least-degree annihilator, as in the hint. Let

$$g = b_0 + b_1 x + \cdots + b_m x^m \neq 0$$

have minimal degree m among all nonzero polynomials with $fg = 0$. Then $b_m \neq 0$. If $m = 0$, then $g = b_0$ is a nonzero constant and $f b_0 = 0$, so we are done. Assume $m \ge 1$. Since the coefficient of x^{n+m} is $a_n b_m$, we have $a_n b_m = 0$. Hence $a_n g$ has degree $< m$ and still annihilates f:

$$f(a_n g) = a_n (fg) = 0.$$

By minimality of m, $a_n g = 0$. Comparing the coefficient of x^{n+m-1} now gives $a_{n-1} b_m = 0$, so $a_{n-1} g$ has degree $< m$ and annihilates f; by minimality, $a_{n-1} g = 0$. Continuing, we obtain

$$a_{n-r} g = 0 \qquad (0 \le r \le n).$$

In particular $a_i b_m = 0$ for every i, hence

$$f b_m = 0.$$

Since $b_m \neq 0$, this is the desired nonzero element annihilating f.

(iii)(b)

Correct: if $af = 0$ with $a \neq 0$, then f is a zero-divisor by definition.

(iv)(⇐): invalid proof of primitivity

Issue. In the proof that f, g primitive implies fg primitive, the displayed formula beginning with "$fg =$" is not actually fg. It is the weighted sum

$$\sum_{i,j} \lambda_i \mu_j b_i a_j x^{i+j}.$$

Expressing 1 as a linear combination of the individual products $a_i b_j$ does not show that 1 lies in the ideal generated by the *coefficients* of fg, which are the anti-diagonal sums

$$c_k = \sum_{i+j=k} a_j b_i.$$

The coefficients $\lambda_i \mu_j$ are not constant on an anti-diagonal, so the terms cannot be regrouped as an A-linear combination of the c_k.

Correction. Reduce modulo a maximal ideal. Let $\mathfrak{m}$ be a maximal ideal of A. In the polynomial ring $(A/\mathfrak{m})[x]$, which is a domain, the image of fg is $\bar{f}\bar{g}$.

If f, g are primitive and fg were not primitive, then there is a maximal ideal $\mathfrak{m}$ containing all coefficients of fg, so $\bar{f}\bar{g} = 0$. Since $(A/\mathfrak{m})[x]$ is a domain, $\bar{f} = 0$ or $\bar{g} = 0$, contradicting the primitivity of f or g.

Conversely, if fg is primitive and f were not primitive, then some maximal ideal $\mathfrak{m}$ contains all coefficients of f; then $\bar{f} = 0$, so $\bar{f}\bar{g} = 0$, contradicting the primitivity of fg. The same argument applies to g. Hence f and g are primitive.

The direction fg primitive $\Rightarrow$ g primitive is also proved by the rearrangement in the original text, provided the last term is corrected from $a_n + \xi_{m+n}$ to $a_n \xi_{m+n}$. The problematic direction is the one discussed above.

Remark after Exercise 2

The remark says "$n + m - 1$ anti-diagonals". The coefficients of fg run from x^0 to x^{n+m}, so there are $n + m + 1$ anti-diagonals, not $n + m - 1$.

Exercise 3

No mathematical error, but the induction is only indicated. For example, for (i) apply Exercise 2(i) with the coefficient ring $R = A[x_1, \ldots, x_{n-1}]$: a polynomial in x_n is a unit in $R[x_n]$ if and only if its constant term is a unit in R and all other coefficients are nilpotent in R. Then apply the induction hypothesis to R. The other parts are analogous.

Exercise 4

Issue. The proof is not merely incomplete; it proves a false statement. In general, $A[x]$ does *not* have Krull dimension 0, and not every prime ideal of $A[x]$ is maximal. For example, if $A = k$ is a field, then

$$(0) \subsetneq (x)$$

are prime ideals of $k[x]$. The phrase "Krull dimension of $\mathfrak{p}$" is also not standard; the relevant condition for $\mathfrak{p}$ to be maximal would be $\dim(A/\mathfrak{p}) = 0$. The citation to Hartshorne, Proposition 1.7, gives $\dim \mathbb{A}^n = n$, not dimension 0.

The correct statement is

$$J(A[x]) = \mathrm{Nil}(A[x]),$$

where J denotes the Jacobson radical and Nil the nilradical.

Correction. Let

$$f = a_0 + a_1x + \cdots + a_nx^n \in J(A[x]).$$

For every $g \in A[x]$, $1 - fg$ is a unit. Taking $g = x$, we see that $1 - xf$ is a unit. By Exercise 2(i), all coefficients of xf are nilpotent; these coefficients are $a_0, a_1, \ldots, a_n$. Hence f is nilpotent. Therefore $J(A[x]) \subseteq \mathrm{Nil}(A[x])$.

Conversely, $\mathrm{Nil}(R) \subseteq J(R)$ for every commutative ring R: if $y^n = 0$, then for every $r \in R$,

$$(1 - yr)(1 + yr + \cdots + (yr)^{n-1}) = 1 - (yr)^n = 1,$$

so $1 - yr$ is a unit, which means $y \in J(R)$. Hence $\mathrm{Nil}(A[x]) \subseteq J(A[x])$, and the two radicals are equal.

The remark that the same holds in $A[x_1, \ldots, x_n]$ is correct, by induction on n.

Exercise 5

(i)(b)

The displayed inverse is missing the factor x^2 in the third term. The beginning of the inverse series should read

$$(a_0 + a_1x + a_2x^2 + \cdots)\left(a_0^{-1} - a_1a_0^{-2}x + (a_1^2a_0^{-3} - a_2a_0^{-2})x^2 + \cdots\right) = 1.$$

More precisely, the coefficients b_n of the inverse are determined recursively by

$$b_0 = a_0^{-1}, \qquad b_n = -a_0^{-1}\sum_{k=1}^{n} a_k b_{n-k} \quad (n \geq 1).$$

(ii)(b): the converse is false

"No" is correct, but a counterexample should be supplied. Let

$$A = \prod_{n \geq 1} k[\varepsilon_n]/(\varepsilon_n^{n+1}),$$

and let $a_n \in A$ be the element whose n-th component is ε_n and whose other components are 0. Then $a_n^{n+1} = 0$, so every a_n is nilpotent. Consider

$$f = \sum_{n \geq 1} a_n x^n \in A[[x]].$$

If $f^N = 0$, then in the n-th component we get $\varepsilon_n^N x^{nN} = 0$. Taking $n > N$, this is impossible because $\varepsilon_n^N \neq 0$ in $k[\varepsilon_n]/(\varepsilon_n^{n+1})$. Hence f is not nilpotent, although all of its coefficients are nilpotent.

(iii)(b): the classification of maximal ideals

Issue. The assertion

"the constant terms of an arbitrary ideal $\mathfrak{I}$ form an ideal I and $\mathfrak{I}_I = \mathfrak{I}$"

is false. For example, in $A[[x]]$, the ideal $\mathfrak{I} = (x^2)$ has constant-term ideal 0, but

$$\mathfrak{I}_0 = \{b_0 + b_1x + \cdots \mid b_0 = 0\} = (x),$$

which is strictly larger than (x^2).

Correction. Let $\mathfrak{m}$ be a maximal ideal of $A[[x]]$, and let

$$\pi : A[[x]] \longrightarrow A, \qquad \pi\Big(\sum a_n x^n\Big) = a_0$$

be the constant-term map, with kernel (x). First we show that $x \in \mathfrak{m}$. If $x \notin \mathfrak{m}$, then x is a unit modulo $\mathfrak{m}$, so there exists $g \in A[[x]]$ with $1 - xg \in \mathfrak{m}$. But $1 - xg$ has constant term 1, hence is a unit in $A[[x]]$, so $\mathfrak{m} = A[[x]]$, contradiction. Thus $x \in \mathfrak{m}$.

Since $x \in \mathfrak{m}$, we have

$$\mathfrak{m} = \pi^{-1}(\pi(\mathfrak{m})).$$

Indeed, if $\pi(f) \in \pi(\mathfrak{m})$, then $f - \pi(f) \in (x) \subseteq \mathfrak{m}$, so $f \in \mathfrak{m}$. Therefore

$$A[[x]]/\mathfrak{m} \cong A/\pi(\mathfrak{m}).$$

Since $\mathfrak{m}$ is maximal, $\pi(\mathfrak{m})$ is a maximal ideal of A. Thus every maximal ideal of $A[[x]]$ is of the form

$$\mathfrak{m} = \pi^{-1}(I) = I + (x) = \{b_0 + b_1 x + \cdots \mid b_0 \in I\}$$

with I maximal in A. Conversely, if I is maximal in A, then $I + (x)$ is maximal because its quotient is A/I. This also proves part (iv): the contraction of $\mathfrak{m}$ is $\pi(\mathfrak{m}) = \mathfrak{m} \cap A$, and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x.

(iii)(a)

With the classification above, the equivalence is immediate. If $a_0 \in J(A)$, then for every maximal ideal I of A, $a_0 \in I$, hence $f \in I + (x)$, which is an arbitrary maximal ideal of $A[[x]]$. Thus $f \in J(A[[x]])$. Conversely, if $f \in J(A[[x]])$, then $f \in I + (x)$ for every maximal $I \subseteq A$, so $a_0 \in I$ for every maximal I, i.e. $a_0 \in J(A)$.

(iv)

This is proved in the correction to (iii)(b).

(v)

The idea in the original text is correct but unfinished. For a prime ideal $\mathfrak{p} \subseteq A$, the ideal

$$\mathfrak{p}A[[x]] = \mathfrak{p}[[x]] = \{a_0 + a_1 x + \cdots \mid a_i \in \mathfrak{p}\}$$

is prime, because

$$A[[x]]/\mathfrak{p}[[x]] \cong (A/\mathfrak{p})[[x]],$$

and $(A/\mathfrak{p})[[x]]$ is a domain. Its contraction to A is $\mathfrak{p}[[x]] \cap A = \mathfrak{p}$. Hence every prime ideal of A is the contraction of a prime ideal of $A[[x]]$. (Alternatively, $\mathfrak{p} + (x)$ is prime with the same contraction.)

Exercise 6

Issue. The proof is unfinished and its logic is incorrect. The statement "if the nilradical equals the Jacobson radical, then every prime ideal contains some maximal ideal" is false: a prime ideal can contain the intersection of all maximal ideals without containing any of them, e.g. (0) in $k[x]$. The claimed equivalence with "every prime ideal is maximal" is also false in general.

Correction. Let J be the Jacobson radical and N the nilradical. Always $N \subseteq J$. Suppose $J \not\subseteq N$, and choose $x \in J \setminus N$. Then the principal ideal (x) is not contained in N, so by hypothesis (x) contains a nonzero idempotent e. Since $x \in J$, we have $e \in J$. But $e(1 - e) = 0$, and $1 - e$ is a unit because $e \in J$; hence $e = 0$, contradiction. Therefore $J \subseteq N$, and so $J = N$.

Exercise 7

Correct. Minor wording: "$A/\mathfrak{p}$ is a integral domain" should be "an integral domain". The hypothesis $n > 1$ is used only to ensure that $n - 1 > 0$; the argument works for every $n \geq 2$.

Exercise 8

Issue. The nilradical is the intersection of all prime ideals, but it need not itself be a prime ideal. Hence it is not an element of the set of prime ideals and cannot be used as a minimal element without further argument. For example, in $A = k[x, y]/(xy)$,

$$(x) \quad \text{and} \quad (y)$$

are minimal prime ideals, while

$$\mathrm{Nil}(A) = (x) \cap (y) = (xy) = (0)$$

is not prime.

Correction. Use Zorn's lemma. The set of prime ideals is nonempty because $A \neq 0$ has a maximal ideal, and maximal ideals are prime. Let $\{\mathfrak{p}_i\}_{i \in I}$ be a chain of prime ideals, and put $\mathfrak{p} = \bigcap_i \mathfrak{p}_i$. We show that $\mathfrak{p}$ is prime. Let $xy \in \mathfrak{p}$ and $x \notin \mathfrak{p}$. Choose i_0 with $x \notin \mathfrak{p}_{i_0}$. For any j: if $\mathfrak{p}_{i_0} \subseteq \mathfrak{p}_j$, then $xy \in \mathfrak{p}_j$ and $x \notin \mathfrak{p}_j$, so $y \in \mathfrak{p}_j$; if $\mathfrak{p}_j \subseteq \mathfrak{p}_{i_0}$, then $y \notin \mathfrak{p}_j$ would imply $x \in \mathfrak{p}_j \subseteq \mathfrak{p}_{i_0}$, contradiction, so again $y \in \mathfrak{p}_j$. Hence $y \in \mathfrak{p}$, and $\mathfrak{p}$ is prime. Zorn's lemma now gives a minimal prime ideal.

Exercise 9

The sketch is correct, but two details should be mentioned. If $\mathfrak{a} = \mathrm{Rad}(\mathfrak{a})$, then Proposition 1.14 gives

$$\mathfrak{a} = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p},$$

an intersection of prime ideals. Conversely, an intersection of prime ideals is radical, since each prime is radical and an intersection of radical ideals is radical. For $\mathfrak{a} = (1)$, the family of primes containing $\mathfrak{a}$ is empty, and the intersection over the empty family is A by convention.

Exercise 10

The proof is essentially correct, but several justifications are missing.

(i) $(i) \Rightarrow (ii)$: If A has exactly one prime ideal $\mathfrak{p}$, then $\mathfrak{p}$ is the unique maximal ideal and $\mathrm{Nil}(A) = \mathfrak{p}$. Every non-unit lies in a maximal ideal, hence in $\mathfrak{p} = \mathrm{Nil}(A)$, so it is nilpotent. Every element outside $\mathfrak{p}$ is a unit. The original text invokes Corollary 1.5 but omits the crucial equality $\mathrm{Nil}(A) = \mathfrak{p}$.

(ii) $(ii) \Rightarrow (iii)$: If $\bar{a} \neq 0$ in $A/\mathrm{Nil}(A)$, then $a \notin \mathrm{Nil}(A)$, so a is not nilpotent. By (ii), a is a unit, hence $\bar{a}$ is a unit. The phrase "$\exists a^{-1} \in A \setminus \mathfrak{R}$" should be replaced by "a is a unit".

(iii) $(iii) \Rightarrow (i)$: If $A/\mathrm{Nil}(A)$ is a field, then $\mathrm{Nil}(A)$ is a maximal ideal. Every prime ideal contains $\mathrm{Nil}(A)$, and since $\mathrm{Nil}(A)$ is maximal, the only prime ideal is $\mathrm{Nil}(A)$ itself. The original conclusion is correct, but one should identify the unique prime as $\mathrm{Nil}(A)$.

Exercise 11

(i)

The chain should start with the Boolean identity for $2x$:

$$(2x)^2 = 2x,$$

while also

$$(2x)^2 = 4x^2 = 4x.$$

Hence $2x = 4x$, so $2x = 0$. As written, "$2x = 4x^2$" is not justified.

(ii)

Correct.

(iii)

The two-generator case is correct. For the induction, let $I = (x_1, \ldots, x_n)$ and set

$$z = 1 - \prod_{i=1}^{n}(1 + x_i).$$

In a Boolean ring, $1 + x_i$ is idempotent because $(1 + x_i)^2 = 1 + 2x_i + x_i^2 = 1 + x_i$. Hence $e = \prod_{i=1}^{n}(1 + x_i)$ is idempotent, and $z = 1 - e$ is idempotent. Moreover, $ex_i = 0$, since the factor $1 + x_i$ times x_i is $x_i + x_i^2 = 2x_i = 0$. Thus

$$zx_i = (1 - e)x_i = x_i,$$

so $x_i \in (z)$ for every i. Therefore $(z) = I$.

Exercise 12

Correct. A zero-divisor cannot be a unit, and in a local ring every non-unit lies in the unique maximal ideal. If $e \neq 0, 1$ is idempotent, then $e(1-e) = 0$, so both e and $1-e$ are zero-divisors, hence non-units, hence both lie in the maximal ideal; then $1 = e + (1 - e)$ lies in the maximal ideal, contradiction. Only the grammar "both e and $1 - e$ is" should be corrected to "are".

Exercise 13

Issue. The proof is not valid as written. The notation $F \in A[f(x_f)]$ is unclear: F should be a polynomial in new variables y_f, evaluated at $y_f = f(x_f)$. The "highest degree" cancellation must be done with the coefficient of the highest power of f_0; since f_0 is monic, $f_0(x_{f_0})^{\alpha_0}$ is not a zero-divisor in the polynomial ring over the other variables. The equation

$$f(x_f)^{\alpha_0}(\text{rest}) = 0$$

is not the correct formulation of "the highest-degree part vanishes". Finally, the termination argument needs the fact that the original relation has no constant term in the y_f variables (or a minimality assumption); it is not spelled out. The parenthetical remark about the characteristic of K is unnecessary and misleading: the argument works in every characteristic.

Correction. Suppose for contradiction that $\mathfrak{a} = (1)$. Then there are finitely many $f_1, \ldots, f_r \in \Sigma$ and $g_1, \ldots, g_r \in A$ such that

$$\sum_{i=1}^{r} g_i f_i(x_{f_i}) = 1.$$

Let L be a field extension of K containing a root α_i of each f_i; for instance, take a splitting field of the product $f_1 \cdots f_r$ over K. Evaluate the identity at

$$x_{f_i} \longmapsto \alpha_i \quad (1 \le i \le r), \qquad x_f \longmapsto 0 \quad (f \notin \{f_1, \ldots, f_r\}).$$

Then every $f_i(x_{f_i})$ becomes $f_i(\alpha_i) = 0$, so the left-hand side is 0, while the right-hand side is 1, contradiction. Hence $\mathfrak{a} \neq (1)$.

Exercise 14

(i)

Zorn's lemma applies, but one should verify that the union of a chain of ideals in Σ again lies in Σ. This is immediate: if every element of every ideal in the chain is a zero-divisor, then every element of the union is a zero-divisor.

(ii): invalid proof that a maximal element is prime

Issue. From $ab \in I$ and $u \neq 0$ with $(ab)u = 0$, one gets $a(bu) = 0$. This only shows that a is a zero-divisor if $bu \neq 0$; if $bu = 0$, it shows that b is a zero-divisor instead. More importantly, even if a is a zero-divisor, maximality of I among ideals consisting of zero-divisors does *not* imply $a \in I$. For example, in $A = k[x, y]/(xy)$, the ideal $I = (x)$ is maximal in Σ, but y is a zero-divisor and $y \notin I$.

Correction. Let I be maximal in Σ, and suppose $ab \in I$. Assume $a \notin I$ and $b \notin I$. By maximality, $I + (a) \notin \Sigma$ and $I + (b) \notin \Sigma$. Hence there exist $i + ra \in I + (a)$ and $j + sb \in I + (b)$ which are not zero-divisors. Their product

$$(i + ra)(j + sb) = ij + isb + jra + rsab$$

lies in I, because $i, j, ab \in I$. Thus the product is a zero-divisor. But the product of two non-zero-divisors is a non-zero-divisor: if $(i + ra)(j + sb)z = 0$, then $(j + sb)z = 0$, and then $z = 0$. Contradiction. Therefore $a \in I$ or $b \in I$, so I is prime.

(iii)

Correct. To justify that every zero-divisor lies in some maximal element of Σ: if x is a zero-divisor with $xu = 0$ and $u \neq 0$, then every element rx of the principal ideal (x) satisfies $(rx)u = 0$, so $(x) \in \Sigma$. By Zorn's lemma, (x) is contained in a maximal element of Σ. Hence the union of all maximal elements of Σ is exactly the set of zero-divisors.

Exercise 15

The answer "Trivial" with references is not a proof. Here are the standard verifications.

(i) $V(E) = V((E))$ by definition. Also $V(\mathfrak{a}) = V(\mathrm{Rad}(\mathfrak{a}))$: a prime ideal contains $\mathfrak{a}$ if and only if it contains $\mathrm{Rad}(\mathfrak{a})$.

(ii) $V(0) = X$, since every prime ideal contains 0; and $V(1) = \varnothing$, since no prime ideal contains 1.

(iii) For a family $(E_i)_{i \in I}$,

$$\mathfrak{p} \in V\Big(\bigcup_i E_i\Big) \iff E_i \subseteq \mathfrak{p} \text{ for all } i \iff \mathfrak{p} \in \bigcap_i V(E_i).$$

Hence $V(\bigcup_i E_i) = \bigcap_i V(E_i)$.

(iv) For ideals $\mathfrak{a}, \mathfrak{b}$,

$$\mathfrak{ab} \subseteq \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}, \mathfrak{b},$$

so

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \subseteq V(\mathfrak{a} \cap \mathfrak{b}) \subseteq V(\mathfrak{ab}).$$

Conversely, if $\mathfrak{p} \supseteq \mathfrak{ab}$ is prime, then $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, so $\mathfrak{p} \in V(\mathfrak{a}) \cup V(\mathfrak{b})$. Hence all three sets are equal.

Exercise 16

(i)

$$\operatorname{Spec} \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ is prime}\}.$$

The zero ideal is missing in the original.

(ii) $\operatorname{Spec} \mathbb{R} = \{(0)\}$. Correct.

(iii)

$$\operatorname{Spec} \mathbb{C}[x] = \{(0)\} \cup \{(x - a) \mid a \in \mathbb{C}\}.$$

The zero ideal is missing in the original.

(iv)

$$\operatorname{Spec} \mathbb{R}[x] = \{(0)\} \cup \{(p) \mid p \in \mathbb{R}[x] \text{ is monic irreducible of degree } \geq 1\}.$$

Over $\mathbb{R}$, the nonconstant irreducibles are the linear polynomials and the quadratics with negative discriminant. The zero ideal is missing in the original.

(v) A complete description is

$$\begin{aligned} \operatorname{Spec} \mathbb{Z}[x] = {} & \{(0)\} \cup \{(p) \mid p \text{ prime}\} \\ & \cup \{(p, f) \mid p \text{ prime},\ f \in \mathbb{Z}[x] \text{ monic and irreducible mod } p\} \\ & \cup \{(g) \mid g \in \mathbb{Z}[x] \text{ primitive and irreducible over } \mathbb{Q}\}. \end{aligned}$$

Here (p) is prime but not maximal; the maximal ideals are exactly the ideals (p, f). The ideal (0) is the generic point, the ideals (p) and (g) are height-one primes, and the ideals (p, f) are the height-two (closed) points.

Exercise 17

Basis

The sets $X_f = X \setminus V(f)$ are open by definition. If $U = X \setminus V(E)$ is an arbitrary open set, then

$$U = X \setminus V(E) = X \setminus \bigcap_{f \in E} V(f) = \bigcup_{f \in E} (X \setminus V(f)) = \bigcup_{f \in E} X_f.$$

Hence the X_f form a basis.

(i)–(iv)

(i)
$$X_f \cap X_g = X \setminus (V(f) \cup V(g)) = X \setminus V(fg) = X_{fg}.$$

(ii) $X_f = \varnothing \iff V(f) = X \iff f \in \mathrm{Nil}(A)$.

(iii) $X_f = X \iff V(f) = \varnothing \iff f$ is a unit.

(iv)
$$X_f = X_g \iff V(f) = V(g) \iff \mathrm{Rad}((f)) = \mathrm{Rad}((g)).$$

(v)

The original argument is essentially correct, but its quantifier notation is unclear. A clean version is: an open cover of X may be refined to a cover by basic open sets X_{f_i}. If $\bigcup_i X_{f_i} = X$, then $\bigcap_i V(f_i) = \varnothing$, so the ideal $\mathfrak{a} = (f_i)$ is not contained in any prime ideal, hence $\mathfrak{a} = (1)$. Therefore there are finitely many indices $i_1, \ldots, i_n$ and elements $g_{i_j} \in A$ with

$$1 = \sum_{j=1}^{n} g_{i_j} f_{i_j}.$$

Then $\bigcup_{j=1}^{n} X_{f_{i_j}} = X$. Thus X is quasi-compact.

(vi): radical, not the ideal itself

Issue. The condition for X_f to be covered by the X_{f_i} is

$$f \in \mathrm{Rad}((f_i)_{i \in I}),$$

not $f \in (f_i)_{i \in I}$. The latter is strictly stronger and is false in general. For example, in $A = k[x]$, take $f = x$ and $f_1 = x^2$. Then $X_f = X_{x^2}$, so X_f is covered by the single set X_{x^2}, but $x \notin (x^2)$.

Correction. Suppose $X_f \subseteq \bigcup_{i \in I} X_{f_i}$. Then

$$V(f) \supseteq \bigcap_{i \in I} V(f_i) = V(\mathfrak{a}),$$

where $\mathfrak{a} = (f_i)$. Hence $f \in \mathrm{Rad}(\mathfrak{a})$, so $f^n \in \mathfrak{a}$ for some n. Since $\mathfrak{a}$ is generated by the f_i, there is a finite subset $J \subseteq I$ with $f^n \in (f_i)_{i \in J}$. Then

$$V(f) = V(f^n) \supseteq V\big((f_i)_{i \in J}\big) = \bigcap_{i \in J} V(f_i),$$

so the finitely many X_{f_i} with $i \in J$ cover X_f. Thus X_f is quasi-compact.

(vii): the compactness argument is wrong

Issue. The negation of "X_0 is a finite union of basic open sets" is not "the index family $\mathcal{F}$ is infinite". An infinite family can contain a finite subcover; for example, X itself is covered by all the X_f, and it is quasi-compact. The attempted contradiction therefore does not work. In addition, the quantifiers in the original only range over countable sequences $X_{f_1}, X_{f_2}, \ldots$, whereas an open cover can be an arbitrary family.

Correction. Let $X_0 \subseteq X$ be open and quasi-compact. Since the X_f form a basis, write

$$X_0 = \bigcup_{f \in \mathcal{F}} X_f$$

for some family $\mathcal{F}$. This is an open cover of X_0. By quasi-compactness, finitely many of these sets cover X_0, so X_0 is a finite union of basic open sets. Conversely, a finite union of quasi-compact spaces is quasi-compact, and each X_f is quasi-compact by (vi).

Exercise 18

(i)–(iii)

Correct. In (i), $\{x\}$ is closed if and only if $V(\mathfrak{p}_x) = \{\mathfrak{p}_x\}$, which is equivalent to $\mathfrak{p}_x$ being maximal. In (iii),

$$y \in \overline{\{x\}} \iff \mathfrak{p}_y \in V(\mathfrak{p}_x) \iff \mathfrak{p}_x \subseteq \mathfrak{p}_y.$$

(iv): $V(\mathfrak{p}_x)$ is not a neighbourhood of x

Issue. The set $V(\mathfrak{p}_x) = \overline{\{x\}}$ is closed, but it is not necessarily a neighbourhood of x: it need not contain an open set containing x. The notation "$V(x)$" is also undefined; the intended set is $V(\mathfrak{p}_x)$. The sentence "I'll interpret neighborhood as closed neighborhood" is not the standard definition and leads to an incorrect proof.

Correction. Let $x \neq y$ and suppose both $y \in \overline{\{x\}}$ and $x \in \overline{\{y\}}$. By (iii), $\mathfrak{p}_x \subseteq \mathfrak{p}_y$ and $\mathfrak{p}_y \subseteq \mathfrak{p}_x$, hence $\mathfrak{p}_x = \mathfrak{p}_y$, so $x = y$, contradiction. Therefore at least one of

$$y \notin \overline{\{x\}}, \qquad x \notin \overline{\{y\}}$$

holds. If $y \notin \overline{\{x\}}$, then $X \setminus \overline{\{x\}}$ is an open neighbourhood of y which does not contain x. Otherwise, $X \setminus \overline{\{y\}}$ is an open neighbourhood of x which does not contain y. This is precisely the T_0 property.

Exercise 19

Issue. The proof is not a proof of irreducibility. In the category of ideals under inclusion, the initial object is the zero ideal (0), not the nilradical. The nilradical is the intersection of all prime ideals; it need not be prime, and it is not an element of $\mathrm{Spec}(A)$ unless it is prime. The expression "$\mathrm{Nil}(A) \subseteq \mathrm{Spec}(A)$" is a type error; the correct condition is $\mathrm{Nil}(A) \in \mathrm{Spec}(A)$, i.e. $\mathrm{Nil}(A)$ is prime. The "connected tree" metaphor does not establish the topological irreducibility of $\mathrm{Spec}(A)$.

Correction. A topological space X is irreducible if and only if every two nonempty open subsets intersect. Since the X_f form a basis, it is enough to check basic open sets. Recall that

$$X_f \neq \varnothing \iff f \notin \mathrm{Nil}(A).$$

Suppose $\mathrm{Nil}(A)$ is prime. If X_f and X_g are nonempty, then $f, g \notin \mathrm{Nil}(A)$, so $fg \notin \mathrm{Nil}(A)$ because $\mathrm{Nil}(A)$ is prime. Hence

$$X_f \cap X_g = X_{fg} \neq \varnothing.$$

Thus every two nonempty basic open sets intersect, so $\mathrm{Spec}(A)$ is irreducible.

Conversely, suppose $\mathrm{Spec}(A)$ is irreducible. If $fg \in \mathrm{Nil}(A)$, then

$$X_f \cap X_g = X_{fg} = \varnothing.$$

By irreducibility, one of X_f, X_g is empty, so $f \in \mathrm{Nil}(A)$ or $g \in \mathrm{Nil}(A)$. Hence $\mathrm{Nil}(A)$ is prime.

Exercise 20

This exercise is skipped in the original. It is not mathematically wrong, but it is incomplete. A brief outline:

(i) If $Y \subseteq X$ is irreducible, then $\overline{Y}$ is irreducible: if U, V are nonempty open subsets of $\overline{Y}$, then $U \cap Y$ and $V \cap Y$ are nonempty open subsets of Y, so they meet, and hence $U \cap V \neq \varnothing$.

(ii) Apply Zorn's lemma to the set of irreducible subspaces of X ordered by inclusion. The union of a chain of irreducible subspaces is irreducible, so maximal elements exist.

(iii) A maximal irreducible subspace is closed, because its closure is irreducible and contains it. The maximal irreducible subspaces cover X, since every singleton is irreducible and is contained in a maximal one. In a Hausdorff space, every irreducible subspace is a singleton: two distinct points have disjoint open neighbourhoods. Hence the irreducible components of a Hausdorff space are its points.

(iv) For $X = \mathrm{Spec}(A)$, the irreducible components are the closed sets $V(\mathfrak{p})$ with $\mathfrak{p}$ a minimal prime ideal. Indeed, $V(\mathfrak{p}) \cong \mathrm{Spec}(A/\mathfrak{p})$ is irreducible when $\mathfrak{p}$ is prime, and every irreducible closed subset is of the form $V(\mathfrak{p})$ with $\mathfrak{p}$ prime. Minimality corresponds to minimality of $\mathfrak{p}$.

Exercise 21

(i)

Correct, apart from the typo $Y_{\phi_{(f)}}$ in the last line, which should be $Y_{\phi(f)}$.

(ii)

The conclusion is correct, but the displayed equality

$$\phi(\phi^*(\mathfrak{q})) = \mathfrak{q}$$

is false in general. What is true is

$$\phi(\phi^{-1}(\mathfrak{q})) \subseteq \mathfrak{q}.$$

The equivalence

$$\mathfrak{a} \subseteq \phi^{-1}(\mathfrak{q}) \iff \phi(\mathfrak{a}) \subseteq \mathfrak{q}$$

is correct. For example, if $\phi : k[x] \to k[x, y]$ is the inclusion and $\mathfrak{q} = (x, y)$, then $\phi^{-1}(\mathfrak{q}) = (x)$ and $\phi((x)) = (x) \neq (x, y)$.

(iii)

Only the easy inclusion is proved in the original; the reverse inclusion is missing. Here is the full proof.

We first show $\phi^*(V(\mathfrak{b})) \subseteq V(\mathfrak{b}^c)$. If $\mathfrak{q} \supseteq \mathfrak{b}$, then

$$\phi^{-1}(\mathfrak{b}) \subseteq \phi^{-1}(\mathfrak{q}) = \phi^*(\mathfrak{q}),$$

so $\phi^*(\mathfrak{q}) \in V(\mathfrak{b}^c)$. Since $V(\mathfrak{b}^c)$ is closed,

$$\overline{\phi^*(V(\mathfrak{b}))} \subseteq V(\mathfrak{b}^c).$$

For the reverse inclusion, let $\mathfrak{p} \in V(\mathfrak{b}^c)$. Suppose $\mathfrak{p} \notin \overline{\phi^*(V(\mathfrak{b}))}$. Then there is a basic open neighbourhood X_f of $\mathfrak{p}$ disjoint from $\phi^*(V(\mathfrak{b}))$, with $f \notin \mathfrak{p}$. For every $\mathfrak{q} \in V(\mathfrak{b})$, this means $\phi^*(\mathfrak{q}) \notin X_f$, i.e.

$$f \in \phi^{-1}(\mathfrak{q}),$$

so $\phi(f) \in \mathfrak{q}$. Thus $\phi(f)$ lies in every prime ideal containing $\mathfrak{b}$, hence

$$\phi(f) \in \mathrm{Rad}(\mathfrak{b}).$$

Therefore

$$f \in \phi^{-1}(\mathrm{Rad}(\mathfrak{b})) = \mathrm{Rad}(\phi^{-1}(\mathfrak{b})) = \mathrm{Rad}(\mathfrak{b}^c).$$

Since $\mathfrak{b}^c \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is prime, $\mathrm{Rad}(\mathfrak{b}^c) \subseteq \mathfrak{p}$, so $f \in \mathfrak{p}$, contradiction. Hence $\mathfrak{p} \in \overline{\phi^*(V(\mathfrak{b}))}$, and the reverse inclusion holds.

(iv)

The original proof is too vague. Here is a precise argument.

If ϕ is surjective, then ϕ^* is injective: if $\phi^*(\mathfrak{q}_1) = \phi^*(\mathfrak{q}_2)$, then

$$\mathfrak{q}_1 = \phi(\phi^{-1}(\mathfrak{q}_1)) = \phi(\phi^*(\mathfrak{q}_1)) = \phi(\phi^*(\mathfrak{q}_2)) = \phi(\phi^{-1}(\mathfrak{q}_2)) = \mathfrak{q}_2.$$

Its image is $V(\ker\phi)$: on one hand, $\ker\phi \subseteq \phi^{-1}(\mathfrak{q})$ for every $\mathfrak{q}$; on the other hand, if $\mathfrak{p} \supseteq \ker\phi$, then $\mathfrak{p} = \phi^{-1}(\phi(\mathfrak{p}))$ and $\phi(\mathfrak{p})$ is prime in $B \cong A/\ker\phi$, so $\mathfrak{p} = \phi^*(\phi(\mathfrak{p}))$.

Finally, ϕ^* is a closed map onto $V(\ker\phi)$: for a closed set $V(\mathfrak{b}) \subseteq Y$, the argument in (iii) together with surjectivity gives

$$\phi^*(V(\mathfrak{b})) = V(\mathfrak{b}^c).$$

Hence ϕ^* is a continuous bijection which is closed, so it is a homeomorphism onto $V(\ker\phi)$. In particular, taking $B = A/\mathfrak{N}$ and $\phi : A \to A/\mathfrak{N}$ the quotient map gives $\mathrm{Spec}(A) \cong \mathrm{Spec}(A/\mathfrak{N})$.

(v)

The chain of equivalences in the original is confused. The correct proof is short. By (iii) with $\mathfrak{b} = 0$,

$$\overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(0^c) = V(\ker\phi).$$

Therefore $\phi^*(Y)$ is dense in X if and only if

$$V(\ker\phi) = X \iff \ker\phi \subseteq \mathrm{Nil}(A).$$

This is the "more precisely" statement.

The condition

$$\mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B))) = \mathrm{Nil}(A)$$

is in fact equivalent to $\ker\phi \subseteq \mathrm{Nil}(A)$, but this equivalence needs proof. Since $\ker\phi \subseteq \phi^{-1}(\mathrm{Nil}(B))$ always, and $\mathrm{Nil}(A) \subseteq \phi^{-1}(\mathrm{Nil}(B))$ always, one has

$$\mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B))) = \mathrm{Nil}(A) \iff \phi^{-1}(\mathrm{Nil}(B)) \subseteq \mathrm{Nil}(A) \iff \ker\phi \subseteq \mathrm{Nil}(A).$$

The original final step only proves the trivial inclusion $\ker\phi \subseteq \phi^{-1}(\mathrm{Nil}(B))$, so it does not establish the desired equivalence.

(vi)

Correct: for $\mathfrak{q} \in \mathrm{Spec}(C)$,

$$(\psi \circ \phi)^*(\mathfrak{q}) = (\psi \circ \phi)^{-1}(\mathfrak{q}) = \phi^{-1}(\psi^{-1}(\mathfrak{q})) = \phi^*(\psi^*(\mathfrak{q})).$$

(vii)

This part is left blank in the original. Here is a solution outline.

Let A be an integral domain with exactly one nonzero prime ideal $\mathfrak{p}$, and let $K = \mathrm{Frac}(A)$. Put

$$B = (A/\mathfrak{p}) \times K, \qquad \phi : A \longrightarrow B, \qquad \phi(x) = (\bar{x}, x).$$

Since $\mathfrak{p}$ is the only nonzero prime, $\mathrm{Spec}(A) = \{(0), \mathfrak{p}\}$, and $A/\mathfrak{p}$ is a field. The spectrum of the product B is the disjoint union

$$\mathrm{Spec}(B) = \{\mathfrak{q}_1, \mathfrak{q}_2\},$$

where

$$\mathfrak{q}_1 = (0) \times K, \qquad \mathfrak{q}_2 = (A/\mathfrak{p}) \times (0).$$

Then

$$\phi^*(\mathfrak{q}_1) = \phi^{-1}(\mathfrak{q}_1) = \mathfrak{p}, \qquad \phi^*(\mathfrak{q}_2) = \phi^{-1}(\mathfrak{q}_2) = (0).$$

Thus ϕ^* is bijective.

However, $\mathrm{Spec}(B)$ is discrete, while $\mathrm{Spec}(A)$ is not: in $\mathrm{Spec}(A)$, the set $\{(0)\}$ is open, but $\{\mathfrak{p}\}$ is not open. Indeed, $X_f = \{(0)\}$ for $f \in \mathfrak{p} \setminus \{0\}$, whereas $f \notin \mathfrak{p}$ gives $X_f = X$, and $f = 0$ gives $X_f = \varnothing$. Hence no basic open set equals $\{\mathfrak{p}\}$. Therefore ϕ^* is a continuous bijection which is not open, so it is not a homeomorphism.

Exercise 22

(a)

Correct. The X_k are clopen, and the map $\mathrm{Spec}(A_k) \to X_k$ is a homeomorphism; this should be stated explicitly.

(b)(I) $(ii) \Rightarrow (i)$

The idea is correct, but one should add that X_1 and X_2 are nonempty, disjoint, open and closed, and that $X_1 \cup X_2 = X$. Hence X is disconnected.

(b)(II) $(iii) \Rightarrow (ii)$

There is a typo: the hypothesis is $e^2 = e$, not $e^2 = 2$. If $e \neq 0, 1$ is idempotent, then

$$A \longrightarrow eA \times (1-e)A, \qquad a \longmapsto (ea, (1-e)a)$$

is an isomorphism. Both factors are nonzero because $e \neq 0$ and $1 - e \neq 0$.

(b)(III) $(i) \Rightarrow (iii)$

The construction is essentially correct, but several details should be justified. If $V(I) \cap V(J) = \varnothing$, then $I + J = (1)$, so there are $a \in I$, $b \in J$ with $a + b = 1$. Since $V(IJ) = X$, we have $IJ \subseteq \mathrm{Nil}(A)$, so $(ab)^n = 0$ for some n. Define

$$s_1 = \sum_{k=0}^{n-1} \binom{2n}{k} a^{2n-k} b^k, \qquad s_2 = \sum_{k=n+1}^{2n} \binom{2n}{k} a^{2n-k} b^k.$$

Then the missing middle term is $\binom{2n}{n}(ab)^n = 0$, so

$$s_1 + s_2 = (a+b)^{2n} = 1.$$

Also $s_1 s_2 = 0$, because every term of the product is divisible by $(ab)^{n+1} = 0$. Hence $s_1(1 - s_1) = 0$, so s_1 is idempotent, and $s_2 = 1 - s_1$. The idempotents are nontrivial: $s_1 \in I$ and I is proper, so $s_1 \neq 1$; $s_2 \in J$ and J is proper, so $s_2 \neq 1$; and $s_1 = 0$ would give $s_2 = 1 \in J$, contradiction. Thus $s_1 \neq 0, 1$.

Exercise 23

(i)

Correct. For every f,

$$V(f) \cap V(1-f) = V((f) + (1-f)) = V(1) = \varnothing,$$

and

$$V(f) \cup V(1-f) = V((f)(1-f)) = V(0) = X,$$

since $f(1-f) = 0$. Hence $V(f) = X_{1-f}$ is open, so X_f is closed. The restriction to nonzero non-unital f is unnecessary.

(ii)

Correct. In a Boolean ring, every finitely generated ideal is principal by Exercise 11(iii), so

$$\bigcap_{k=1}^{n} V(f_k) = V\Big(\sum_{k=1}^{n}(f_k)\Big) = V((f))$$

for some f, hence

$$X_{f_1} \cup \cdots \cup X_{f_n} = X_f.$$

(iii): the clopen-set argument is wrong

Issue. The claim that one can choose $f \in \mathfrak{p}$ which lies in no other prime of Y is not Proposition 1.11(i), and it is false in general when Y is infinite. Moreover, even if such an f exists, the fact that $V(f)$ is a clopen set containing $\mathfrak{p}$ does not imply $V(f) = Y$; there can be many clopen sets containing $\mathfrak{p}$. The sentence "by $f \in \mathfrak{p}, \mathfrak{p} \in Y$ we have $V(f) = Y$" is unjustified.

Correction. Let $Y \subseteq X$ be clopen. Since X is quasi-compact by Exercise 17(v), and Y is closed, Y is quasi-compact. Since Y is open, it is a union of basic open sets:

$$Y = \bigcup_{f \in \mathcal{F}} X_f.$$

By quasi-compactness, finitely many of these cover Y, say $X_{f_1}, \ldots, X_{f_n}$. By (ii), there is an $f \in A$ with

$$X_{f_1} \cup \cdots \cup X_{f_n} = X_f.$$

Hence $Y = X_f$. Conversely, every X_f is clopen by (i). Therefore the clopen subsets of X are exactly the sets X_f.

(iv): the Hausdorff proof is wrong

Issue. The sets

$$V_1 = X \setminus \{\mathfrak{q}\}, \qquad V_2 = X \setminus \{\mathfrak{p}\}$$

are *open*, not closed, because $\{\mathfrak{p}\}$ and $\{\mathfrak{q}\}$ are closed (Exercise 18(i)). The reference to Exercise 17(iv) should also be Exercise 17(v).

Correction. First, X is quasi-compact by Exercise 17(v). To prove that X is Hausdorff, let $\mathfrak{p} \neq \mathfrak{q}$ be distinct points. Since they are maximal ideals (Exercise 11(ii)), $\mathfrak{p} + \mathfrak{q} = A$, so there are $a \in \mathfrak{p}$, $b \in \mathfrak{q}$ with $a + b = 1$. In a Boolean ring, a and b are idempotents, and $b = 1 - a$. Then

$$X_a = \{\mathfrak{r} \in X \mid a \notin \mathfrak{r}\}, \qquad X_b = \{\mathfrak{r} \in X \mid b \notin \mathfrak{r}\}$$

are disjoint clopen sets, with $\mathfrak{q} \in X_a$ (because $a \notin \mathfrak{q}$) and $\mathfrak{p} \in X_b$ (because $b \notin \mathfrak{p}$). Hence X is Hausdorff.

Exercise 24

(a)(I)(iv): the associativity computation is wrong

Issue. The displayed computation

$$(a+b)+c = (ab' \vee a'b)c \vee (ab + a'b')c$$

does not correctly expand $(a+b)+c$, and the final expression is not symmetric in a, b, c.

Correction. Recall $x + y = (xy') \vee (x'y)$. Then

$$\begin{aligned}(a+b)+c &= \big((ab' \vee a'b)c'\big) \vee \big((ab' \vee a'b)'c\big) \\ &= (ab'c' \vee a'bc') \vee \big((a' \vee b)(a \vee b')c\big) \\ &= ab'c' \vee a'bc' \vee a'b'c \vee abc.\end{aligned}$$

This expression is symmetric in a, b, c (it is the set of terms with an odd number of unprimed variables), so addition is associative.

(a)(II)(iii)

"Associativity of multiplication is given by the associativity of $\vee$" should be "of $\wedge$", since multiplication is $a \wedge b$.

(b)(iii): the absorption computation is wrong

Issue. The line

$$a \vee (a \wedge b) = a + ab + aab = 3a$$

is incorrect; $a + ab + aab = a + 2ab$, not $3a$.

Correction. Since $a^2 = a$ and $2ab = 0$,

$$a \vee (a \wedge b) = a \vee (ab) = a + ab + a^2 b = a + ab + ab = a + 2ab = a.$$

The other absorption law follows by duality.

Missing details in (b)

To show that A is a Boolean lattice with respect to $a \leq b \iff a = ab$, one should also verify that $\leq$ is a partial order (reflexivity, antisymmetry, transitivity), that $a \wedge a' = 0$ (in addition to $a \vee a' = 1$), and that complements are unique. These are short verifications. For example,

$$a \wedge a' = a(1-a) = a - a^2 = 0,$$

and if b is another complement of a, then

$$b = b \wedge 1 = b \wedge (a \vee a') = (b \wedge a) \vee (b \wedge a') = 0 \vee (b \wedge a') = b \wedge a',$$

so $b \leq a'$; similarly $a' \leq b$, hence $b = a'$.

Exercise 25

The statement is the conclusion of Stone's theorem, not a proof. The intended argument is:

(i) By Exercise 24, $A(L)$ is a Boolean ring.

(ii) By Exercise 23, the clopen subsets of $X = \mathrm{Spec}(A(L))$ are exactly the sets X_f with $f \in A(L)$.

(iii) The map

$$L \longrightarrow \{\text{clopen subsets of } X\}, \qquad a \longmapsto X_a$$

preserves the Boolean lattice structure (order, sup, inf, complement).

(iv) This map is bijective by Exercise 23(iii).

Hence L is isomorphic to the lattice of clopen subsets of the compact Hausdorff space X.

Exercise 26

Issue. In the first inclusion, the implication

$$\mathfrak{m}_y \in \widetilde{U}_f,\ f \in \mathfrak{m}_y \implies f(y) \neq 0$$

has the wrong sign. By definition,

$$\widetilde{U}_f = \{\mathfrak{m} \in \mathrm{Max}(C(X)) \mid f \notin \mathfrak{m}\}.$$

Thus $\mathfrak{m}_y \in \widetilde{U}_f$ means $f \notin \mathfrak{m}_y$, i.e. $f(y) \neq 0$. Also, the second inclusion is written as $U_f \subseteq \widetilde{U}_f$, but these are subsets of different spaces; it should be $\mu(U_f) \subseteq \widetilde{U}_f$, or equivalently $U_f \subseteq \mu^{-1}(\widetilde{U}_f)$.

Correction. Let $\mathfrak{m} \in \widetilde{U}_f$. By (i), $\mathfrak{m} = \mathfrak{m}_y$ for some $y \in X$. Since $\mathfrak{m}_y \in \widetilde{U}_f$, we have $f \notin \mathfrak{m}_y$, hence $f(y) \neq 0$, so $y \in U_f$ and $\mathfrak{m} = \mu(y) \in \mu(U_f)$. Thus $\widetilde{U}_f \subseteq \mu(U_f)$.

Conversely, if $x \in U_f$, then $f(x) \neq 0$, so $f \notin \mathfrak{m}_x$, hence $\mu(x) = \mathfrak{m}_x \in \widetilde{U}_f$. Thus $\mu(U_f) \subseteq \widetilde{U}_f$.

Therefore $\mu(U_f) = \widetilde{U}_f$. Since μ is bijective by (i) and (ii), and the sets U_f (respectively $\widetilde{U}_f$) form a basis of the topology, μ is a homeomorphism.

Exercise 27

Issue. The argument is circular. The step

$$\text{“}\exists x \in X \text{ such that } \varphi^{-1}(\mathfrak{m}) \text{ is the maximal vanishing ideal of } x\text{”}$$

is precisely the weak Nullstellensatz (every maximal ideal of $k[t_1, \ldots, t_n]$ is of the form $(t_1 - x_1, \ldots, t_n - x_n)$). This is not available at this point in the book; the text itself says that surjectivity of μ is one form of Hilbert's Nullstellensatz, proved in Chapter 7. There is also a notation error: $\varphi : X \to X/I(X)$ should be $\varphi : k[t_1, \ldots, t_n] \to P(X)$.

Correction. The correct statement at this point is the equivalence: μ is surjective if and only if every maximal ideal of $P(X)$ is a vanishing ideal $\mathfrak{m}_x$. Under the quotient map $\varphi : k[t_1, \ldots, t_n] \to P(X)$, maximal ideals of $P(X)$ correspond to maximal ideals of $k[t_1, \ldots, t_n]$ containing $I(X)$. Thus surjectivity of μ is equivalent to the weak Nullstellensatz. A proof should either cite Chapter 7 or prove the Nullstellensatz; the argument in the original file assumes it.

Exercise 28

Issue. The proof omits the crucial verification that the map

$$\varphi = (F_1, \ldots, F_m) : X \longrightarrow k^m$$

actually lands in Y. It is not enough to say that $(F_1, \dots, F_m)$ "defines a regular mapping $X \to Y$"; one must show that for every $x \in X$, the point $(F_1(x), \dots, F_m(x))$ satisfies all the equations defining Y. The proof also does not prove injectivity of the correspondence.

Correction. Let $\Phi : P(Y) \to P(X)$ be a k-algebra homomorphism, and let ξ_i be the coordinate functions on Y. Put

$$F_i = \Phi(\xi_i) \in P(X) \qquad (1 \le i \le m).$$

For $g \in I(Y)$, we have $g(\xi_1, \dots, \xi_m) = 0$ in $P(Y)$, hence

$$0 = \Phi(g(\xi_1, \dots, \xi_m)) = g(F_1, \dots, F_m)$$

in $P(X)$. Therefore, for every $x \in X$,

$$g(F_1(x), \dots, F_m(x)) = 0.$$

This shows that $(F_1(x), \dots, F_m(x)) \in Y$, so $\varphi = (F_1, \dots, F_m)$ is a regular map $X \to Y$.

Let $\varphi^* : P(Y) \to P(X)$ be the induced homomorphism $\eta \mapsto \eta \circ \varphi$. On the generators ξ_i,

$$\varphi^*(\xi_i) = \xi_i \circ \varphi = F_i = \Phi(\xi_i).$$

Since $P(Y)$ is generated as a k-algebra by $\xi_1, \dots, \xi_m$, it follows that $\varphi^* = \Phi$. Hence the correspondence is surjective.

For injectivity, suppose $\varphi, \psi : X \to Y$ induce the same k-algebra homomorphism. Then for every i,

$$\xi_i \circ \varphi = \xi_i \circ \psi,$$

so the i-th coordinates of $\varphi(x)$ and $\psi(x)$ agree for every $x \in X$. Hence $\varphi = \psi$. Therefore the correspondence is bijective.

Unfinished or sketchy exercises

The following exercises are not necessarily wrong, but they are incomplete:

- **Ex. 3**: the induction is only indicated.
- **Ex. 5(v)**: the idea is correct; see the correction above.
- **Ex. 9**: the proof is only a reference to Proposition 1.14.
- **Ex. 15**: "Trivial" with references is not a proof.
- **Ex. 16(v)**: $\operatorname{Spec} \mathbb{Z}[x]$ is left unfinished.
- **Ex. 17**: the basis and parts (i)–(iv) are asserted without proof; (vi) and (vii) contain errors.
- **Ex. 20**: skipped.
- **Ex. 21(iii),(v),(vii)**: (iii) and (v) are incomplete; (vii) is blank.
- **Ex. 25**: only the statement of Stone's theorem is given.
- **Ex. 27**: the argument is conditional on the Nullstellensatz.
- **Ex. 28**: the key verification and injectivity are missing.

Minor typographical corrections

- Ex. 1: replace $1 - x^r$ by $1 - (-x)^r$ in the intermediate identity.
- Ex. 2(i)(b): $a_n s^n$ should be $a_n x^n$.
- Ex. 2, remark: "$n + m - 1$ anti-diagonals" should be "$n + m + 1$ anti-diagonals".
- Ex. 5(i)(b): the third term of the inverse series should contain x^2.
- Ex. 10: the nilradical is usually denoted $\mathfrak{N}$, not $\mathfrak{R}$; "a integral domain" should be "an integral domain".
- Ex. 11(i): the equality $2x = 4x^2$ needs the justification $(2x)^2 = 2x$.
- Ex. 21(i): $Y_{\phi_{(f)}}$ should be $Y_{\phi(f)}$.
- Ex. 21(ii): $\phi(\phi^{-1}(\mathfrak{q})) = \mathfrak{q}$ should be $\phi(\phi^{-1}(\mathfrak{q})) \subseteq \mathfrak{q}$.
- Ex. 22(b)(II): $e^2 = 2$ should be $e^2 = e$.
- Ex. 23(iv): "Ex-17(iv)" should be "Ex-17(v)".
- Ex. 24(a)(II)(iii): "associativity of $\vee$" should be "associativity of $\wedge$".
- Ex. 24(b)(iii): "$= 3a$" should be "$= a + 2ab = a$".
- Ex. 26: $f \in \mathfrak{m}_y$ should be $f \notin \mathfrak{m}_y$; and $U_f \subseteq \widetilde{U}_f$ should be $\mu(U_f) \subseteq \widetilde{U}_f$.
- Ex. 27: $\varphi : X \to X/I(X)$ should be $\varphi : k[t_1, \ldots, t_n] \to P(X)$.
- Ex. 28: the diagram should have $P(Y) = k[t_1, \ldots, t_m]/I(Y)$; "For any k-algebra homomorphism ϕ" should be "Φ"; and "regualr" should be "regular".